Market Lens

Release market intelligence for Indian music labels - project summary

Prepared 23 October 2026 | Status: working pilot on a development laptop (not yet deployed to the server)

6 labels tracked	**76** official YouTube channels	**1,435** songs, 15 Sep - 15 Oct	**1.34B** YouTube views (these songs)
13,966 YouTube videos collected	**1,399** songs with singer credits	**466** songs matched on Spotify	**622** songs with Apple metadata

Summary

Market Lens collects public data about new song releases from the official YouTube channels of Sony Music India, T-Series, Zee Music Company, Saregama, Aditya Music and Think Music. It groups each song's uploads (video, lyric, audio, Shorts, teasers) into one song, adds metadata from Spotify, Apple Music, MusicBrainz and the credits in the video descriptions, and records views, likes and comments over time. Matching, merging and metrics use fixed, explainable rules. Everything is run, reviewed and explored in a web console (Market Lens Console). An AI Insights page shows rule-based insight panels with a written AI-assist assessment; live AI is built and ready, but switched off until the LLM integration.

The pilot started as a Spotify-first exercise for Sony Music India (Phase 1 of the Times Music Phase 0 plan), moved to a YouTube-first design when Spotify's developer limits proved restrictive, and was then extended to six labels, full song metadata, duplicate merging, cross-platform lookups and insight panels.

1. What was built

Area	What it does
Song discovery (YouTube-first)	Collects uploads from 76 official channels of 6 labels; videos published in the song window are grouped into songs. Shorts and teasers join a song but never create one.
One song, many versions	Each song keeps every upload with its own views, likes and comments, plus totals by type (video, lyric, audio, Shorts, teaser).
Spotify metadata	One search per song (ISRC, album, release date, artists); extra Spotify copies of the same recording are linked at no extra cost; an artist-first route finds new releases through artists' release lists while search is blocked.
Other metadata sources	Apple iTunes Search API (album, release date, artists, genre), MusicBrainz (ISRCs), Deezer (rank by ISRC).
Song metadata	Parsed from the credits blocks in video descriptions: singers, music director, lyricist, cast, film, album, director, producer, arranger, mixing, choreographer, label; plus rule-based language, category and occasion. Every value records its source.
Duplicate merging	Four configurable rules (same ISRC, same Spotify title + artist, spelling variant, contained name) that suggest or auto-merge; bulk merge in one click; every merge is undoable and survives later runs.
Review and decision log	Uncertain matches go to a Review queue; every approve, reject, link and merge is logged with who made it and can be undone exactly.
Explore	Metrics for every metadata value: songs, new releases, views, share, day-on-day gain, likes, comments, average per song, top song, daily gains and collaborators.
AI Insights	Rule-based panels (emerging hits, cross-label, catalog opportunities, artist portfolio, anomalies), a per-song comparison with similar songs, and a written AI-assist briefing.
Outputs	Per-run CSVs for songs (with metadata and per-type views), YouTube videos and song metadata (one row per value, for Power BI).

Console pages

Page	Purpose
Overview	KPIs, songs per day, songs by label and channel, top songs, biggest day-on-day gainers
Songs	All songs with filters (label, channel, language, category, status, date range), credits search, multi-select merge, CSV export
Explore	Leaderboards per metadata field and a detail view per artist, film, language and more
AI Insights	Insight panels with AI-assist notes; Ask Music AI marked as coming with the API integration
Review	Spotify links to check, duplicate songs, video matches, decision history with undo
Runs / Outputs	Start or cancel runs, live log, run history; every CSV and raw API response
Sources / Settings	API keys and quota per key; release window, labels and channels, merge rules, schedule

2. Labels, channels and data collected

Channels were chosen the same way for every label: the channels a label's main YouTube channel features on its own page, keeping music channels and leaving out movie, TV-show, kids, karaoke, remix and old-song re-upload channels and look-alike channels.

Label	Channels	Songs in window	Notes
T-Series	25	619	Main, 14 regional, 10 devotional / classical
Saregama	19	280	Main, 12 language channels, devotional, ghazal, sufi, classical
Sony Music India	8	204	Main, VEVO, South, Malayalam, Bangla, Bhakti, Classical
Aditya Music	7	163	Main, Hindi, Tamil, Kannada, Malayalam, Bhakthi (2)
Zee Music Company	12	140	Main, South, 8 regional, Devotional, Ibadat
Think Music	5	27	India, Telugu, Malayalam, Think Indie, Think Divine

Song window: songs first uploaded 15 September to 15 October 2026 (configurable). Uploads are collected from 31 days before the window so earlier versions are linked too.

Coverage of the 1,435 songs

Item	Songs	Source
Language	1,433	Title words, YouTube audio language, channel, title script
Singers / music director	1,399 / 1,398	YouTube credits, Spotify and Apple artists
Lyricist	1,244	YouTube credits
Film	704	Credits, Spotify, Apple Music, titles
Spotify match (new 189 / older 277)	466	Spotify search and artist-first lookups
Apple Music metadata	622	iTunes Search API
ISRC	472	Spotify, MusicBrainz
Deezer rank	501	Deezer track by ISRC
Spotify lookup still pending	~775	Resumes automatically when Spotify's daily block lifts

Review work done

- 28 review decisions recorded (Spotify links approved or rejected, duplicate songs merged or marked different), all undoable.
- Bugs found during review and fixed: Indian-script vowel signs dropped from names; merges failing when songs had metadata; generic title parts ('Durga Puja Original Song') taken as song names; renamed songs losing their Spotify link.

3. Early insights (AI assist, 23 October 2026)

Rules compute every number; the notes below are a written assessment of those numbers. Only two days of daily readings exist (16 and 22 Oct), so growth and acceleration signals are early.

Label	Songs	Median percentile vs songs of the same age	Share above typical
Think Music	27	85	96%
Sony Music India	204	57	56%
Saregama	280	57	58%
Aditya Music	163	48	48%
Zee Music Company	140	46	44%
T-Series	619	43	45%

- **Volume is not performance.** T-Series has 43% of songs and the most views (577M), but its typical song is in the 43rd percentile; many uploads are small regional and devotional releases.
- **Language:** Hindi (66th) and Haryanvi (62nd) over-perform; Kannada (20th), Gujarati (28th) and Punjabi (31st) trail.
- **Strong pockets:** T-Series Haryanvi (91st), Think Music Tamil and Telugu (85th-87th), Zee Gujarati (85th), T-Series Bengali (84th).
- **Catalogue matters:** 32% of songs were first released a year or more earlier; e.g. Nagada Sang Dhol (2013) and Lallati Bhandar (2013) are among the fastest uploads of the month.
- **Engagement check:** 20 of the 300 fastest songs get under 1 like per 1,000 views (typical about 6); six of the eight lowest are Zee releases. This can indicate paid promotion (hypothesis).
- **Emerging hits** (not yet top 5%): Awadhiya Ke Raja (Zee), Those Pe Badmashi (T-Series), Humsafar (Saregama), Abar Elo Maa (T-Series), Chitta Kukkad (T-Series): 8-17 days old, 190,000-390,000 views a day.

AI status

Live AI (LLM explanations and an 'Ask Music AI' chat with read-only data tools) is built and switched off by decision; enabling it later needs an LLM key and two endpoints turned on. Today's notes are a dated, manual assessment and do not update with the data.

4. API limits and data constraints

Source	Limit observed / documented	Impact and handling
YouTube Data API	10,000 units/day per Google Cloud project; 11 keys configured, used in order	A daily sync of all 76 channels uses about 300-500 units. Quota resets at 12:30 PM IST.
Spotify (development mode)	About 600 searches/day; about 90 album lookups/day; no popularity, label or play counts; batch endpoints blocked; owner needs Premium	Metadata only. Runs stop cleanly on a quota block and resume automatically. Extended quota (company application) is the lasting fix.
Apple iTunes Search API	About 20 requests/minute; no ISRC	About 50 minutes for 1,000 songs; 63% matched.
MusicBrainz	1 request/second; volunteer data, lags on new Indian releases	About 15-20% of songs get an ISRC, mostly older songs.
Deezer	Search returns nothing from India; ISRC lookups work	Used for rank by ISRC only.
Public data overall	Current totals only; no history before collection starts; no watch time, revenue or demographics	Daily snapshots must run every day; trends build up over time.

Terms of service

Spotify's terms prohibit scraping and restrict commercial analytics and derived metrics; YouTube limits how long API data may be stored without refresh; Apple, Deezer and MusicBrainz have their own terms. Web scraping of Spotify was deliberately not used. A terms-of-service sign-off (requested by Times Music) is needed before commercial use.

Database recommendation

For daily collection from January 2025 across all major Indian labels (estimated 300,000-500,000 videos, about 180M reading rows a year): PostgreSQL 16 with the TimescaleDB extension (compression and automatic daily totals), raw API responses in S3, and Power BI on the daily total tables. A 4 vCPU / 16-32 GB server covers the first 2-3 years. PostgreSQL alone with monthly partitions is a fully open-source alternative. ClickHouse only if readings reach tens of billions of rows.

5. Open items and next steps

#	Item	Why
1	Deploy the laptop version to the server and run it as a service	Daily history only builds while the Console runs every day
2	Apply for Spotify extended quota (as a company)	Development-mode limits block scale
3	Terms-of-service sign-off for YouTube, Spotify, Apple, Deezer and MusicBrainz data	Required before commercial use
4	Move to PostgreSQL + TimescaleDB	Scale, concurrency and a direct Power BI connection
5	Naming and people clean-up	Songs named after singers, duo names split (Ajay / Atul), spelling variants of the same artist
6	Refresh AI-assist notes after 7+ days of daily readings	Growth, acceleration and anomaly signals become reliable
7	Integrate the LLM (when approved)	Live explanations and Ask Music AI
8	Optional: Spotify Charts CSV upload, Last.fm listener counts	The only legitimate Spotify stream counts; extra popularity signal
9	Per-user logins, HTTPS and key rotation	Keys were shared in chat; the Console uses one shared password

Technical notes

- Code: Python 3 (standard library plus the optional Anthropic SDK), SQLite, a single-page web console; about 7,200 lines.
- Key modules: sync_youtube, songs (grouping, Spotify / iTunes / MusicBrainz / Deezer), spotify_artists, match, merge, metadata, insights, review, studio (server and API), ai (switched off).
- Project notes for future work and the README document how to run the Console and how each step works.

Appendix A. API limitations in detail

Limits as observed in Market Lens runs up to 23 October 2026 and as documented by each provider. Spotify does not publish its development-mode quotas; the figures below are what we measured.

A1. YouTube Data API v3 (main source: songs and views)

	Details
Daily quota	10,000 units per Google Cloud project, reset at midnight Pacific time (12:30 PM IST).
Cost per call	videos.list, playlistItems.list, channels.list, channelSections.list: 1 unit per call (up to 50 items). search.list: 100 units, so the pipeline never uses search.
Our usage	A full sync of 76 channels cost 309 units (5,729 videos read); a normal daily sync is about 300-500 units. Finding channels for new labels cost about 1,200 units once.
Keys	11 keys configured, used in order; the next takes over when one runs out. Keys only add quota if they come from different Google Cloud projects.
Available	Views, likes, comment count, title, description (with credits), tags, publish date, duration, audio language, category, channel.
Not available	Watch time, impressions, click-through rate, retention, revenue, demographics, dislikes (these need the channel owner's YouTube Analytics).
Gaps	Current totals only: daily trends exist only from the day collection starts and cannot be backfilled. Likes or comments can be hidden or disabled; subscriber counts are rounded.
Raising the limit	Quota increase through Google's audit form (a compliance review of how the data is used).
Terms	YouTube's developer policies limit how long API data may be stored without refresh; keeping views indefinitely needs the terms-of-service sign-off.

A2. Spotify Web API (metadata) - development mode

	Details
Rate limit	Rolling 30-second window; exceeding it returns 429 with Retry-After, which the code respects.
Daily quota	Separate per endpoint group, not published. Search: blocked after about 617 calls in 24 hours (block about 21 hours). Album and artist lookups: blocked after about 90 calls on 23 Oct and about 600-900 on 16 Oct (block about 23-24 hours).
Restrictions	Search pages capped at 10 results; batch calls (/tracks?ids=, /albums?ids=) blocked (403); popularity, label, artist followers and genres not returned; the app owner needs Premium (done).
Never available	Stream or play counts, in any mode. They come only from Spotify for Artists or distributor reports for a label's own catalogue, or Spotify Charts (top 200 CSV).
Our handling	One search per song, per-run budgets, clean stop and automatic resume on a block, artist-first lookups while search is blocked; iTunes and MusicBrainz fill gaps. Backlog: about 775 songs, 1.5-2 days at about 600 searches a day.
Raising the limit	Extended quota mode via the developer dashboard. Since 2025 Spotify reportedly accepts only organisations with an established business and a large user base (around 250,000 monthly users); confirm current criteria.
Terms	Scraping is prohibited. Commercial analytics, derived listening metrics and AI or ML training on the data are restricted.

A3. Apple iTunes Search API (fallback metadata)

	Details
Limit	About 20 requests a minute, no key; calls spaced 3 seconds apart (about 50 minutes per 1,000 songs).
Available	Album, release date, artists, genre, length, explicit flag, Apple track link (India storefront).
Not available	ISRC, popularity, play counts.

	Details
Our result	622 of 988 songs matched; a random check of 10 matches found all correct.
Terms	Intended for promoting and linking to Apple content; commercial use needs the terms review.

A4. MusicBrainz (open data: ISRCs)

	Details
Limit	1 request a second per IP address (503 above that); an identifying User-Agent is required (ours: MarketLens/0.1).
Coverage	Volunteer-maintained and slow to add new Indian releases: about 15-20% of our songs, mostly older ones.
Speed	About 4 seconds per song (two searches plus spacing), about 1 hour per 1,000 songs.
Terms	Core data is CC0 (public domain, commercial use allowed); heavy commercial users are asked to support MusicBrainz or use its paid data feed.

A5. Deezer API (rank)

	Details
Limit	50 requests per 5 seconds, no key.
Search	Returns no results from India (Deezer is not available in India): a match count but an empty result list.
Works	Track lookup by ISRC, giving Deezer's rank (an undocumented popularity score, not streams). Rank read for 501 songs.
Terms	Commercial use needs Deezer's API terms review.

A6. LLM API (planned, switched off)

	Details
Limits	Rate limits per organisation usage tier; billed per token (the model planned for this costs about $4 per million input tokens and $20 per million output).
Design	Explanations cached once per panel per day to keep cost low; Ask Music AI uses read-only tools over the data.

A7. Constraints across all sources

- **No historical data:** no API gives past daily plays or views; trends exist only from the day collection starts, so the pipeline must run every day without gaps.
- **No real play counts for Spotify or Apple:** only YouTube gives public performance numbers.
- **Collector uptime:** the Console currently runs as a task of a development session on a laptop; daily collection needs it running as a service on the server.
- **Terms of service:** every source has storage and commercial-use terms; the sign-off requested by Times Music is needed before commercial use.

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